Skills Summary

Counting Steps, Finding Ratios, and Knowing When a Sum Exists

Use this reference while completing your worksheet or when studying.

Skill 1 — Count steps, not terms

Arithmetic: $a_n = a_1 + (n-1)d$. **Geometric:** $a_n = a_1 r^{n-1}$.

- The first term costs no steps, so reaching term n takes $n-1$ of them.
- **The $n = 1$ test:** substitute $n = 1$. A correct formula returns a_1 exactly; a formula using nd returns the second term instead.
- Sum of n arithmetic terms: $S_n = \frac{n}{2}(a_1 + a_n)$ — and a_n here must be the nth term, not the one after it.

Example: An arithmetic sequence has $a_1 = 5$ and $d = 3$. Find the 20th term.
Step 1. The formula that jumps straight to a term is the explicit one, and its whole risk is the step count, so write $(n-1)$ before substituting anything.
Step 2. $a_{20} = 5 + (20-1)(3) = 5 + 57$
The 20th term is **62**.

Try it: An arithmetic sequence has $a_1 = 8$ and $d = 6$. Find the 15th term.

check: **92**

Skill 2 — Ratios divide, and only some sums exist

Common ratio: $r = \dfrac{a_{k+1}}{a_k}$ — divide, never subtract, and confirm on a second pair before trusting it.

Infinite sum: $S_\infty = \dfrac{a_1}{1-r}$, valid *only* when $|r| < 1$. The denominator is $1 - r$; dividing by r instead is the most common mis-copy. If $|r| \geq 1$ the series diverges and no sum exists.

Example: Find the sum of the infinite geometric series with $a_1 = 12$ and $r = \frac{1}{4}$.
Step 1. Check convergence first: $\left|\frac{1}{4}\right| < 1$, so the sum exists and the formula may be used at all.
Step 2. $S_\infty = \dfrac{12}{1 - \frac{1}{4}} = \dfrac{12}{\frac{3}{4}}$
The sum is **16**.

Try it: Find the sum of the infinite geometric series with $a_1 = 20$ and $r = \frac{1}{5}$.

check: **25**

Skill 3 — Which k does a binomial term want?

Binomial theorem: $(x+p)^n = \sum \binom{n}{k} x^{n-k} p^k$.

- k counts the *constant's* power, not x's. For the x^m term, solve $n-k=m$, so $k=n-m$.
- The two exponents always add to n — check every term against that.
- $\binom{n}{k} = \binom{n}{n-k}$, so the coefficient alone will not tell you that k is wrong. Only the power of p will.

Example: Find the coefficient of x^3 in $(x+2)^5$.
Step 1. Set the index from the exponent that is missing: $n-k=3$ with $n=5$ forces $k=2$, so the constant is squared, not cubed.
Step 2. $\binom{5}{2} x^3 2^2 = 10 \cdot x^3 \cdot 4$
The coefficient is **40**.

Try it: Find the coefficient of x^4 in $(x+2)^6$.

check: **60**

(!) Watch out: An off-by-one index usually produces a clean, plausible number — that is why it survives. Test at $n=1$, and substitute your answer back into the original sequence before you trust it.